Naming Conventions

This document outlines the naming conventions used for schemas, tables, views, columns, and other objects in the data warehouse.

Table of Contents

1. General Principles
2. Table Naming Conventions
    - Bronze Rules
    - Silver Rules
    - Gold Rules
3. Column Naming Conventions
    - Surrogate Keys
    - Technical Columns
4. Stored Procedure

General Principles

- **Naming Conventions**: Use snake_case, with lowercase letters and underscores (_) to separate words.
- **Language**: Use English for all names.
- **Avoid Reserved Words**: Do not use SQL reserved words as object names.

Table Naming Conventions

Bronze Rules

- All names must start with the source system name, and table names must match their original names without renaming.
- **`<sourcesystem>_<entity>`**
    - `<sourcesystem>`: Name of the source system (e.g., `crm`, `erp`).
    - `<entity>`: Exact table name from the source system.
    - Example: `crm_customer_info` → Customer information from the CRM system.

Silver Rules

- All names must start with the source system name, and table names must match their original names without renaming.
- **`<sourcesystem>_<entity>`**
  - `<sourcesystem>`: Name of the source system (e.g., `crm`, `erp`).
  - `<entity>`: Exact table name from the source system.
  - Example: `crm_customer_info` → Customer information from the CRM system.

Gold Rules

- All names must use meaningful, business-aligned names for tables, starting with the category prefix.
- **`<category>_<entity>`**
  - `<category>`: Describes the role of the table, such as `dim` (dimension) or `fact` (fact table).
  - `<entity>`: Descriptive name of the table, aligned with the business domain (e.g., `customers`, `products`, `sales`).
  - Examples:
    - `dim_customers` → Dimension table for customer data.
    - `fact_sales` → Fact table containing sales transactions.

Glossary of Category Patterns

Pattern	Meaning	Example(s)
`dim_`	Dimension table	`dim_customer`, `dim_product`
`fact_`	Fact table	`fact_sales`
`report_`	Report table	`report_customers`, `report_sales_monthly`

Column Naming Conventions

Surrogate Keys

- All primary keys in dimension tables must use the suffix `_key`.
- **`<table_name>_key`**
  - `<table_name>`: Refers to the name of the table or entity the key belongs to.
  - `_key`: A suffix indicating that this column is a surrogate key.
  - Example: `customer_key` → Surrogate key in the `dim_customers` table.

Technical Columns

- All technical columns must start with the prefix `dwh_`, followed by a descriptive name indicating the column's purpose.
- **`dwh_<column_name>`**
  - `dwh`: Prefix exclusively for system-generated metadata.
  - `<column_name>`: Descriptive name indicating the column's purpose.
  - Example: `dwh_load_date` → System-generated column used to store the date when the record was loaded.

Stored Procedure

- All stored procedures used for loading data must follow the naming pattern:
- **`load_<layer>`**.
  - `<layer>`: Represents the layer being loaded, such as `bronze`, `silver`, or `gold`.
  - Example:
    - `load_bronze` → Stored procedure for loading data into the Bronze layer.
    - `load_silver` → Stored procedure for loading data into the Silver layer.